

Requirements Document

The Library Management Database is designed to store and manage information about books, authors, library members, and borrowing activity. Library employees can use the system to register members, add books, view available books, check books out, process book returns, and review borrowing records. The database provides an organized way to track which books are owned by the library and which members currently have books checked out.

Allows employees to:

- Add new books
- Add authors
- Register new members
- View available books
- Check books out
- Record returned books
- Search for books

System has to store the following (Data requirements):

Members:

- Member ID
- First name
- Last name
- Email
- Phone num
- Membership date

Authors:

- Author ID
- First name
- Last name

Books:

- Book ID
- Title
- ISBN
- Publication yr
- Author ID
- Available copies

Loans:

- Loan ID
- Member ID
- Book ID
- Checkout date

- Due date
- Return date

Use Cases:

Use case 1: Register a new member

User: Library employee

The employee enters the members name, email, phone, and membership date. The system stores the new member in the database

Expected result: A new member record is created

Use case 2: Add a new book

User: Library employee

The employee enters the book title, book ID publication year, number of copies and author

Expected result: A new book record is added to the database

Use case 3: View available books

User: Library employee or member

The user requests a list of books with at least one available copy

Expected result: The system displays all books currently available for checkout

Use case 4: Check out a book

User: Library employee

The employee selects a member and a book. The system creates a loan record and subtracts one from the total available copies

Expected result: The book is assigned to the member and its availability is updated

Use case 5: Return a book

User: Library employee

The employee records the return date for an active loan.

Expected result: The loan is marked as returned and the number of available copies is increased

Use case 6: Borrowing history

User: Library employee

The employee enters a member ID to view the books the member has borrowed

Expected result: The system displays the members previous and current loans